1 **(a)** $8x - 4 = 3x + 11$ **[M1]**

$5x = 15$ **[M1]**

3 **[A1]**

(b) $2y = 3x - 5$ **[M1]**

$2y + 5 = 3x$ **[M1]**

$\frac{2y+5}{3}$ or $\frac{2y}{3} + \frac{5}{3}$ **[A1]**

[Total Marks: 6]

2 **(a)** $5x - 10 = 3x + 14$ **[M1]**

$2x = 24$ **[M1]**

12 **[A1]**

(b) $2A = h(a + b)$ or $\frac{A}{h} = \frac{a+b}{2}$ **[M1]**

$\frac{2A}{h} = a + b$ or $2A = ha + hb$ **[M1]**

$\frac{2A}{h} - a$ or $\frac{2A - ah}{h}$ **[A1]**

[Total Marks: 6]

3 **(a)** Multiply second equation by 2: $4x - 10y = 18$ **[M1]**

Subtract equations to eliminate x: $13y = -13$ **[M1]**

$y = -1$ **[A1]**

$x = 2$ **[A1]**

(b) $(3x + 2)(x - 4) = 0$ **[M2]**

If M0, M1 for $(3x \pm a)(x \pm b)$ where $ab = 8$ or $a \pm 3b = -10$

$x = -\frac{2}{3}$ **[A1]**

$x = 4$ **[A1]**

[Total Marks: 8]

4 **(a)** $2(3x + 2 + x - 1) = 42$ or $3x + 2 + x - 1 = 21$ **[M1]**

$4x + 1 = 21$ **[M1]**

5 **[A1]**

(b) $p^2 = \frac{x-3}{x}$ **[M1]**

$p^2 x = x - 3$ **[M1]**

$p^2 x - x = -3$ or $x - p^2 x = 3$ or $x(p^2 - 1) = -3$ **[M1]**

$\frac{3}{1-p^2}$ or $\frac{-3}{p^2-1}$ **[A1]**

[Total Marks: 7]

5 (a) $\frac{3(x+3)-2(x-1)}{(x-1)(x+3)} = 1$ **[M1]**

$3x+9-2x+2 = (x-1)(x+3)$ **[M1]**

$x+11 = x^2+2x-3$ **[M1]**

$x^2+x-14 = 0$ established with no errors seen **[A1]**

(b) $\frac{-1 \pm \sqrt{1^2-4(1)(-14)}}{2(1)}$ **[M1]**

$\frac{-1 \pm \sqrt{57}}{2}$ **[M1]**

3.27, −4.27 **[A1]**

[Total Marks: 7]

6 (a) $y(x+2) = 3x-1$ **[M1]**

$yx+2y = 3x-1$ **[M1]**

$2y+1 = 3x-yx$ or $yx-3x = -2y-1$ **[M1]**

$x = \frac{2y+1}{3-y}$ or $x = \frac{-2y-1}{y-3}$ **[A1]**

(b) Multiply throughout by $(x-1)(x+2)$: $2(x+2)+3(x-1) = (x-1)(x+2)$ **[M1]**

Expand terms: $2x+4+3x-3 = x^2+x-2$ **[M1]**

Collect to form a quadratic equation: $x^2-4x-3 = 0$ **[A1]**

Use the quadratic formula: $x = \frac{-(-4) \pm \sqrt{(-4)^2-4(1)(-3)}}{2(1)}$ **[M1]**

$x = \frac{4 \pm \sqrt{28}}{2}$ **[M1]**

$x = 4.65$ and $x = -0.65$ **[A1]**

[Total Marks: 10]